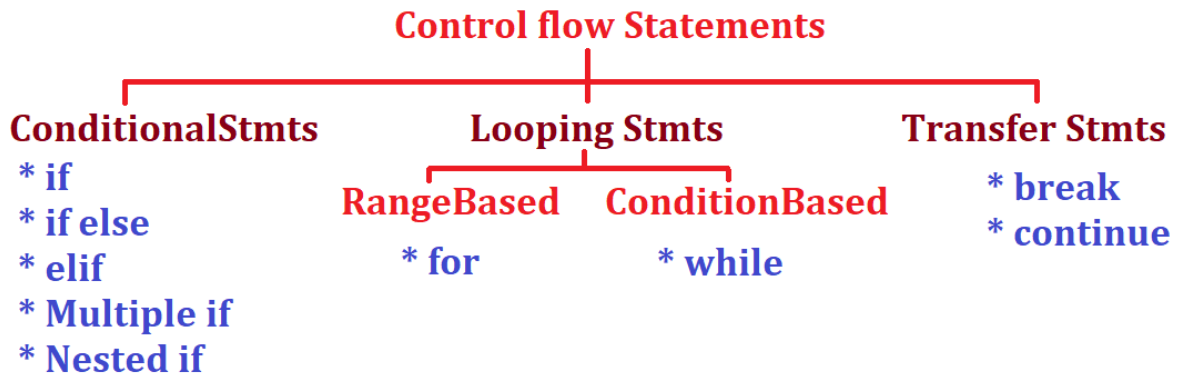


Control flow Statements

- > Control flow statements are used to control flow of execution
- > Control flow statements are 3 types



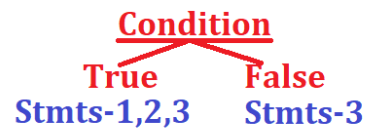
1. if

- > If you want to execute certain statements based on condition then use 'if'

Syntax:

```
if(condition):
    Statements-1
    Statement-2
    Statement-3
```

(indentation) } if-block



Note: If condition is True then only if-block is Executed

Ex:

```
a = int(input('Enter any Number : '))
if(a>=5):
    a = a + 3
print(a)
```

a
7

Testcase1:

Input : 7
Output : 10

Testcase2:

Input : 4
Output : 4

Testcase3:

Input : 5
Output : 8

2. if-else

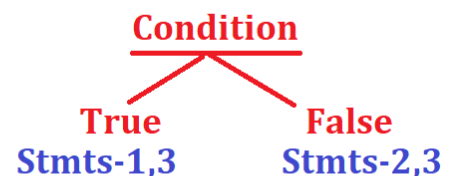
- > If you want to execute only one set out of more than two sets then use if-else

Syntax:

```
if(condition):
    Statement-1
    .....
else:
    Statement-2
    .....
Statement-3
```

} if-block

} else-block



- > First it checks condition, if it is True then it will execute if-block otherwise it will execute else-block

--> W.a.p to check given number is Even number or odd number

```
n = int(input('Enter any Number : '))
if (n%2==0):
    print('Even Number')
else:
    print('Odd Number')
```

--> W.a.p to read age of a person then check person is eligible for vote or not

```
age = int(input('Enter Age : '))
if (age>=18):
    print('Eligible for vote')
else:
    print('Not Eligible for vote')
```

--> W.a.p to read s1,s2,s3 marks, then check student is pass or Fail,
if pass then only calculate total, avg and print total, avg
otherwise print message as 'Better luck next time'

```
s1 = int(input('Enter sub-1 Marks : '))
s2 = int(input('Enter sub-2 Marks : '))
s3 = int(input('Enter sub-3 Marks : '))
if s1>=35 and s2>=35 and s3>=35:
    total = s1 + s2 + s3
    avg = total/3
    print('total is : ', total)
    print('Avg is : ', avg)
else:
    print('Better Luck Next Time')
```

3. elif

--> If you want to execute only one set out of more than two sets then use elif

Syntax:

```
if( condition-1 ):
    Statement-1  } if-block
elif(condition-2):
    Statement-2  } elif-block
else:
    Statement-3  } else-block
Statement-4
```

--> First it checks Condition-1, if it is True then it will execute if-block
Otherwise it checks condition-2, if it is True then it will execute elif-block
Otherwise it will execute else-block

Note:

Python checks conditions one by one
First True condition executes
Remaining conditions are skipped

-->W.a.p to read a number then check number is +ve or -ve or Zero

```
n = int(input('Enter any Number : '))
if n>0:
    print('+ve Number')
elif n<0:
    print('-ve Number')
else:
    print('Zero')
```

--> W.a.p to read 3 numbers then print biggest number

```
a = int(input('Enter any Number : '))
b = int(input('Enter any Number : '))
c = int(input('Enter any Number : '))
if a>b and a>c:
    print('Biggest Number is : ',a)
elif b>c:
    print('Biggest Number is : ',b)
else:
    print('Biggest Number is : ',c)
```

-->W.a.p to read 3 subject marks, then find Total, avg and Grade

avg>=70	---> Grade - A
avg>=60 and avg<70	---> Grade - B
avg>=50 and avg<60	---> Grade - C
Otherwise	---> Grade - D

```
s1 = int(input('Enter Sub-1 Marks : '))
s2 = int(input('Enter Sub-2 Marks : '))
s3 = int(input('Enter Sub-3 Marks : '))
total = s1 + s2 + s3
avg = total / 3
print('Total Marks : ',total)
print('Average is : ',avg)
if avg>=70:
    print('Grade - A')
elif avg>=60 and avg<70:
    print('Grade - B')
elif avg>=50 and avg<60:
    print('Grade - C')
else:
    print('Grade - D')
```